

ULIP Integration Requirement Document

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Project Name: ULIP

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Revision History

Version	Date	Author	Reviewer	Significant Changes
1.0	12/09/2023	Ayush Arya	Nayab Ali	Baseline Version

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Introduction

1.1 Purpose

The purpose of this document is to define and provide details for accessing API for vehicle information from ULIP system and check the given data is valid or not.

This document details the following points:

1. **Data Integration between VAHAN and ULIP web service:** Identification of operational points where data integration between VAHAN and ULIP web service will be needed.
2. **Technical Approach** for performing Data Integration

The Overall Description

User will share the vehicle information such as **vehicle number, chassis number, engine number and owner name** based on data send by user ULIP system will call respective VAHAN APIs for check the entered information is valid or not. In this **vehicle number** is a mandatory field and in **chassis number, engine number and owner name**, we have to enter all or any of these field in request body.

Following are the point for accessing vehicle information: -

1. This API will share vehicle data based on information provided by user.
2. For **AUTHAPI/02** ULIP will take **vehicle number, chassis number, engine number and owner name** from user and get data from vahan API and check the given data is valid or not.

1.2 Data Integration between VAHAN System and ULIP application

User will share information to ULIP system based on data shared ULIP will provide data information to user.

1.3 AUTHAPI/02

This API will take vehicle number and connect with VAHAN API to get data.

S. No.	Field Name	Field Description	Format	Required/Not – Required	Length
1	vehiclenumber	Vehicle Number as available in VAHAN system	^[a-zA-Z0-9]{5,11}\$	Required	Min 5 Max 11
2	chasisnumber	Chassis Number as available in VAHAN system	^[a-zA-Z0-9]{1,20}\$	Not Required	Min 1 Max 20
3	enginenumner	Engine Number as available in VAHAN system	^[a-zA-Z0-9]{1,20}\$	Not Required	Min 1 Max 20
4	ownername	Owner Name as available in Vahan system	^[a-zA-Z0-9\s]*\$	Not Required	

1.3.1 Technical Approach

1. All the data will be shared through rest web services.
2. ULIP system will provide **vehicle number, chasis number, engine number and owner name** for which detail is required.
3. VAHAN API will share details of vehicle which exist in VAHAN system in xml format.
4. Data receive from VAHAN is in XML format. ULIP system convert this data to string format and assign it to key "**response**".
5. ULIP system will provide the data in JSON Object format in response body.

Example:

User will **share Vehicle Number and Chasis number or Engine number or Owner name** to ULIP system, then ULIP system will make a request to VAHAN API.

Request:

The request must be of the following format, where vehicle number is passed within the URL body in JSON format.

For staging -

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/AUTHAPI/02>

```
{
  "vehiclenumber": "UP15AU1291",
  "ownername": "SHADAB ALI",
  "chasisnumber": "MA3EWDE1S00242299",
  "enginenumner": "K10BN7017526"
}
```

Curl URL

```
curl --location --
request POST 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/AUTHAPI/02' \
--
header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJlbGwIiwiaWF0IjoxNjk0Njg
50TEyLCJhcHBzIjoizGF0YXB1c2gifQ.pp4HaM_j5mkHIqa0rb6BUCDEuRNBAXhUWAhIRM12qoSoioUr3IuGdu
qlUMJth-kvLR1dZyan0sbL3Pwp0xOFow' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--header 'Cookie: __zlcmid=1DSlSDwigKpTJa2; _fw_crm_v=b6a11a26-11c3-4b74-d0d1-
3b9c43a5b7c7' \
--data-raw '{
    "vehiclenumber": "UP15AU1291",
    "chasisnumber": "MA3EWDE1S00242299",
    "enginenummer": "K10BN7017526",
    "ownername": "SHADAB ALI"
}'
```

For production -

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/AUTHAPI/02>

```
{
    "vehiclenumber": "UP15AU1291",
    "ownername": "SHADAB ALI",
    "chasisnumber": "MA3EWDE1S00242299",
    "enginenummer": "K10BN7017526"
}
```

Curl URL

```
curl --location --
request POST 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/AUTHAPI/02' \
--
header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJlbGwIiwiaWF0IjoxNjk0Njg
50TEyLCJhcHBzIjoizGF0YXB1c2gifQ.pp4HaM_j5mkHIqa0rb6BUCDEuRNBAXhUWAhIRM12qoSoioUr3IuGdu
qlUMJth-kvLR1dZyan0sbL3Pwp0xOFow' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--header 'Cookie: __zlcmid=1DSlSDwigKpTJa2; _fw_crm_v=b6a11a26-11c3-4b74-d0d1-
3b9c43a5b7c7' \
--data-raw '{
    "vehiclenumber": "UP15AU1291",
    "chasisnumber": "MA3EWDE1S00242299",
    "enginenummer": "K10BN7017526",
}
```

```
"ownername": "SHADAB ALI"  
}
```

For obtaining authentication token, refer [Section 1.4](#)

Response

To acknowledge the above request, ULIP will send the below response -

a) In case of invalid format

```
{  
  "response": null,  
  "error": "true",  
  "code": "400",  
  "message": "Data format failed OR wrong value entered at: vehiclenumber. Format should follow ^[A-Z0-9]{5,11}$"  
}
```

b) In case of chasis number, engine number and owner name does not exist in VAHAN system

```
{  
  "response": [  
    {  
      "response": {  
        "enginenum": "Failed",  
        "chasisnumber": "Failed",  
        "ownername": "Failed"  
      },  
      "responseStatus": "SUCCESS"  
    }  
  ],  
  "error": "false",  
  "code": "200",  
  "message": "Success"  
}
```

c) In case if we do not enter the vehicle number

```
{  
  "response": null,  
  "error": "true",  
  "code": "400",  
  "message": "vehiclenumber:is mandatory field."  
}
```

```
}
```

d) In case if we do not enter the chasis number, engine number and owner name

```
{
  "response": null,
  "error": "true",
  "code": "400",
  "message": "All the input field is missing!Please enter at least one of these: chasisnumber,enginenum,ownername"
}
```

e) In case if we enter wrong vehicle number

```
{
  "response": [
    {
      "response": {
        "errorDescription": "Vehicle Details not Found"
      },
      "responseStatus": "SUCCESS"
    }
  ],
  "error": "false",
  "code": "200",
  "message": "Success"
}
```

f) In case vehicle exist

```
{
  "response": [
    {
      "response": {
        "enginenum": "Verified",
        "chasisnum": "Verified",
        "ownernum": "Verified"
      },
      "responseStatus": "SUCCESS"
    }
  ],
  "error": "false",
  "code": "200",
  "message": "Success"
}
```

Status Code 200: vehicle data response

Status Code 400: Bad request (Invalid json syntax, invalid json key)

Status Code 401 or 403: Unauthenticated or Unauthorized

Status Code 500: Some internal server error occurred

Status Code 502: Server is not responding.

1.4 Data Transmission & Authentication Mechanism

All data exchange would be done over secure HTTP (HTTPS). Request from ULIP's system would be made in VAHAN system through a data exchange URL that has following form:

For staging -

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/AUTHAPI/02>

Body:

```
{
  "vehiclenumber": "UP15AU1291",
  "ownername": "SHADAB ALI",
  "chasisnumber": "MA3EWDE1S00242299",
  "enginenumner": "K10BN7017526"
}
```

Access to above data exchange URL would be authorized by use of a security mechanism implemented by the ULIP system. It is as follows:

- First time access: It would require a username and password [which would be shared with user beforehand]. Once user hit with valid user id and password, ULIP system will return a basic authorization token to user, using that token user will communicate with ULIP system until token will not get expired.

Brief summary of HTTP Basic Authentication is as follows:

First time when User will hit ULIP secured API through username and password for access token as follow-

```
curl --location --request POST 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/user/login' \
--header 'Accept: application/json' \
--header 'Content-Type: application/json' \
--data-raw '{
  "username": "xxxx",
  "password": "xxxx@123"
}'
```

Authorization:**Bearer**

```
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJlbGluIiwiaWF0IjoxNjk0Njg5OTEyLCJhchBzIjoizGF0YXB1c2gifQ.pp4HaM_j5mkHIqa0rb6BUCDEuRNBAxhUWAhIRM12qoSoioUr3IuGduqlUMJth-kvLR1dZyan0sbL3Pwp0xOFow
```

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which represent user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired and user need to re authenticate.

Using The above token, user can request ULIP system for vehicle details as follow -

```
curl --location --
request POST 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/AUTHAPI/02' \
--
header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJlbGluIiwiaWF0IjoxNjk0Njg5OTEyLCJhchBzIjoizGF0YXB1c2gifQ.pp4HaM_j5mkHIqa0rb6BUCDEuRNBAxhUWAhIRM12qoSoioUr3IuGduqlUMJth-kvLR1dZyan0sbL3Pwp0xOFow' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--header 'Cookie: __zlcmid=1DS1SDwigKpTJa2; _fw_crm_v=b6a11a26-11c3-4b74-d0d1-3b9c43a5b7c7' \
--data-raw '{
    "vehiclenumber": "UP15AU1291",
    "chasisnumber": "MA3EWDE1S00242299",
    "enginenummer": "K10BN7017526",
    "ownername": "SHADAB ALI"
}'
```

For Production –

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/user/login' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--data-raw '{
    "username": "xxxx",
    "password": "xxxx@123"
}'
```

Authorization:**Bearer**

eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGluIiwiaWF0IjoxNzI3MTYxMDgwLCJhcHBzIjoizGF0YXB1c2gifQ.XhVZm7NiCD05-v9091WJQhYzdk1a0Shs_Ley0T4R2MX63swRyxHbHxh6bVXtzqWwWiSLwwNGRQIcqGhr49hpGg

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which represent user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired and user need to re authenticate.

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<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/AUTHAPI/02>

```
curl --location --  
request POST 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/AUTHAPI/02' \  
--  
header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGluIiwiaWF0IjoxNzI3MTYxMDgwLCJhcHBzIjoizGF0YXB1c2gifQ.pp4HaM_j5mkHIqa0rb6BUCDEuRNBAXhUWAhIRM12qoSoioUr3IuGduqlUMJth-kvLR1dZyan0sbL3Pwp0xOFow' \  
--header 'Content-Type: application/json' \  
--header 'Accept: application/json' \  
--header 'Cookie: __zlcmid=1DS1SDwigKpTJa2; _fw_crm_v=b6a11a26-11c3-4b74-d0d1-3b9c43a5b7c7' \  
--data-raw '{  
    "vehiclenumber": "UP15AU1291",  
    "chasisnumber": "MA3EWDE1S00242299",  
    "enginenumber": "K10BN7017526",  
    "ownername": "SHADAB ALI"  
}'
```